# Library Management System - MoSCoW Prioritization

*Scores: Impact and Feasibility rated 1 (low) to 5 (high). Priority Score = Impact x Feasibility (formula). Category is the final judgement, informed by the score and by dependencies.*

| # | Requirement | Impact (1-5) | Impact on Users & Stakeholders | Feasibility (1-5) | Feasibility (Time / Budget / Resources) | Priority Score | MoSCoW Category | Justification |
|---|---|---|---|---|---|---|---|---|
| 1 | **Search books by title and author** | 5 | Every member and librarian uses it daily; without it the catalogue is unusable. | 5 | Standard database query (SQL LIKE / full-text index); low cost, a few days of work. | 25 | **Must-Have** | Core function of any library system. Very h effort, and other features (reservation, QR, |
| 2 | **Online book reservation system** | 5 | Members avoid wasted visits and long queues; librarians can manage demand for popular titles. | 4 | Needs availability status, a reservation queue and expiry rules; moderate effort on a standard web stack. | 20 | **Must-Have** | Central to the 'online' goal of the project and benefit. Feasible within the timeline, so it be release. |
| 3 | **User-friendly dashboard for librarians** | 5 | Librarians run issue, return, reservation and overdue tasks from one screen; saves time and reduces errors. | 4 | Front-end work over existing data; moderate effort, no extra licences needed. | 20 | **Must-Have** | Primary interface for the main operational s adoption of the whole system suffers. |
| 4 | **Monthly reports on borrowed books for administrators** | 4 | Gives administrators data for purchasing, stock and usage decisions; not needed for daily operations. | 4 | Aggregate queries plus PDF/Excel export; low-moderate effort once borrowing data exists. | 16 | **Should-Have** | Important to administrators but the system s follow right after the core release, and repor in the meantime. |
| 5 | **Email notifications for overdue books** | 4 | Improves on-time returns and fine collection; reduces manual follow-up by librarians. | 4 | Uses SMTP or a free-tier mail service plus a scheduled job; low cost, moderate effort. | 16 | **Should-Have** | High value and feasible, but librarians can c manually at first. Needs borrowing and due- before it can be built. |

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| 6 | **QR code scanning for borrowing and returning** | 3 | Faster checkout and fewer typing errors, but manual entry or barcode lookup already works. | 3 | Needs QR generation, labelling of every book and scanner/camera integration; extra hardware and labelling effort. | 9 | **Could-Have** | A convenience that speeds up the process Worth adding if time and budget remain afte Have items. |
| 7 | **Allow users to review and rate books** | 2 | Nice for engagement and discovery; does not affect core library operations. | 4 | Simple form and rating table; low effort, but needs moderation to avoid abuse. | 8 | **Could-Have** | Easy to build but low business impact. Inclu capacity exists. |
| 8 | **Multi-language support** | 3 | Valuable where members read in several languages, but not essential if one primary language is used. | 3 | Requires translating all interface text and a localization framework; effort grows with each language. | 9 | **Could-Have** | Useful for accessibility and inclusion, but th in one language. Best planned for after laun stable. |
| 9 | **Integrate a chatbot for user assistance** | 2 | Answers common questions, but a help page, FAQ and librarian support already cover most needs. | 2 | Needs NLP/AI service integration, training content and ongoing maintenance; high effort and running cost for the benefit. | 4 | **Won't-Have** | Low impact relative to cost and complexity. release, where an FAQ page serves as the |
| 10 | **Develop a mobile app version of the system** | 3 | Convenient for members, but a mobile-friendly web interface gives most of the same benefit. | 1 | Separate Android/iOS builds, app-store accounts, extra testing and maintenance; exceeds the time, budget and team size. | 3 | **Won't-Have** | Not feasible within current constraints. Won responsive web design covers mobile users, be revisited later. |